Quantifying the Damage: A Data-Driven Evaluation of COVID-19 Mortality and Healthcare Strain

Group: OAT
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The COVID-19 pandemic necessitated data-driven analysis to understand the virus's severity and intervention effectiveness. Using the Our World in Data (OWID) COVID-19 dataset, including a curated US subset, we analyzed infection rates, hospitalizations, and deaths. This analysis offers a comprehensive view of the pandemic's progression in the US, focusing on infection severity, hospital strain, and mortality.

Analysis of ICU Rate and Case Fatality Rate

Building on the dataset and group objectives, this section focuses on two key severity indicators: the icu_rate, which measures the share of COVID-19 cases requiring intensive-care hospitalization, and case_fatality_rate, which reflects the proportion of confirmed cases that resulted in death. Examining these features together provides insight into how critical illness and mortality evolved in the pandemic and what factors may have influenced their patterns.

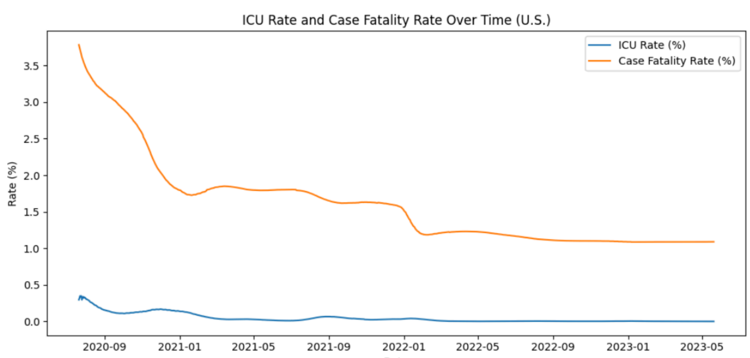


Figure 1. ICU Rate (%) and Case Fatality Rate over time in the United States, showing early-pandemic peaks and a gradual decline after 2021.

Above we see ICU Rate & Case Fatality Rate in this time series time plot correlate with a steady decline across the timeline. CFR is around 3 to 4% in 2020 and drops below 1% by 2023, ICU Rate starts around 0.5%. The high early values correspond to the initial pandemic waves, when severe cases and deaths were more common. Which accounts for the early disparity, where severe and underreported cases were in the initial phase. Possible reasons being because no vaccine had been made available yet and hospitals were still figuring out better treatments.

Correlation and Regression Analysis

To assess whether new infections and hospital admissions could explain changes and get more insight in ICU Rate, a correlation and regression analysis was done. Shown in Figure 2, correlation coefficients show weak associations between ICU Rate and both 7-day-average new cases (r equal to about 0.14), and weekly hospital admissions (r equal to about 0.35). A Ridge regression model using these predictors yielded an R^2 (coefficient of determination) of -1478.1, indicating that the predictors explained virtually none of the variation in ICU Rate and an RMSE (Root Mean Squared Error) of 0.041, indicating that predictions deviated from the true ICU Rate values by an average of roughly 4 percentage points.

Although this magnitude of error is relatively small, the highly negative R-squared suggests that the model's predictions did not follow the actual data trend, meaning that ICU Rate variation cannot be explained by the selected predictors. Confirming that short-term cases of hospitalization volumes do not linearly predict ICU occupancy rates. Figure 3 further supports this finding, as the wide scatter of points illustrates little correlation between hospitalization levels and ICU percentages. These results suggest that ICU demand was shaped primarily by external factors such as vaccination coverage, virus variant severity, and improvement in clinical care, rather than by the raw number of infections or hospitalizations observed in the dataset.

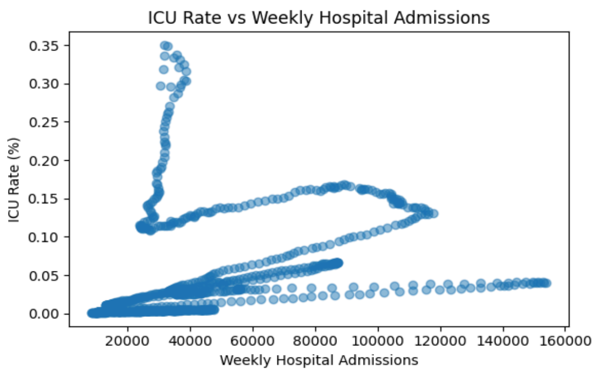


Figure 3. Scatter plot showing the weak linear relationship between ICU Rate and weekly hospital admissions.

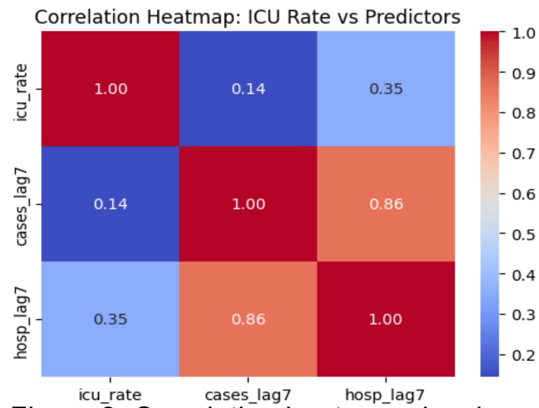


Figure 2. Correlation heatmap showing relationships between ICU Rate, lagged new Cases, and weekly hospital admissions.

Outlier Detection

Next we take a look at a box plot that was used to identify potential outliers in the ICU Rate using the Interquartile Range (IQR) method. Several high-value points above approximately 0.12 (12%) were flagged as statistical outliers. However, taking a closer look these values correspond to legitimate pandemic peak periods rather than data entry errors. This shows a highly right-skewed plot, with most days showing low ICU occupancy and a few extreme periods of intense healthcare strain. This shows that ICU demand spiked sharply during early pandemic waves but remained low in later stages as vaccinations rolled out and there was better medical capacity.

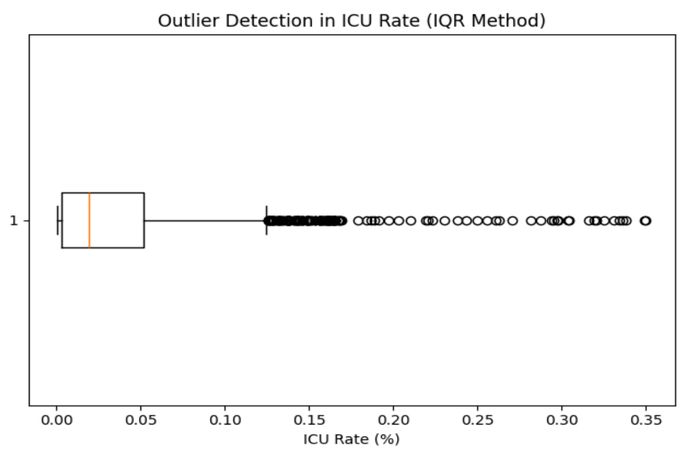


Figure 4. Box plot showing ICU Rate (%) distribution with statistical outliers identified using IQR.

Relationship Between ICU Rate and Case Fatality Rate

As we saw the relationship between ICU Rate and Case Fatality Rate provides a strong indicator of overall COVID-19 severity in the U.S. Correlation analysis revealed a very strong positive relationship (r = 0.869), meaning that periods with higher ICU occupancy were also those with higher fatality proportions. This association remained consistent even when fatality rates were lagged by two weeks, aligning with the expected delay between critical illness and reported deaths. Over time, both ICU Rate and CFR declined sharply, reflecting improved treatments, possibly expanded ICU capacity, and the rollout of vaccinations.

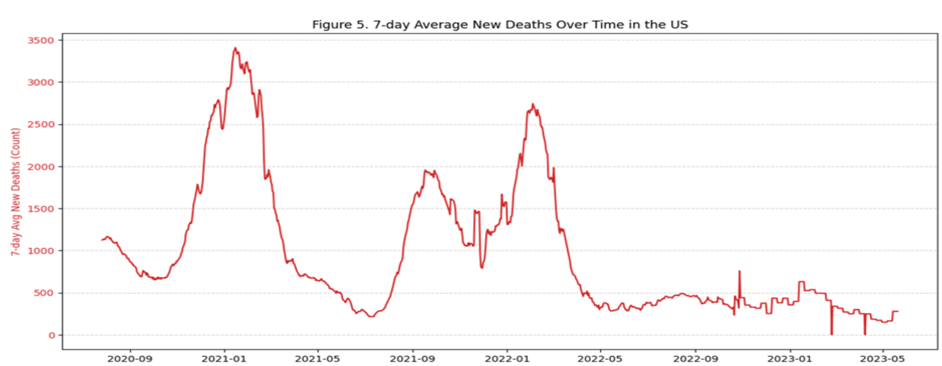
```
                        case_fatality_rate  icu_rate  7day_avg_new_cases  \
case_fatality_rate                   1.000     0.869              -0.054
icu_rate                             0.869     1.000               0.144
7day_avg_new_cases                  -0.054     0.144               1.000
weekly_hosp_admissions               0.089     0.345               0.862

                        weekly_hosp_admissions
case_fatality_rate                       0.089
icu_rate                                 0.345
7day_avg_new_cases                       0.862
weekly_hosp_admissions                   1.000
                    icu_rate  case_fatality_rate  cfr_lag14
icu_rate               1.000               0.869      0.875
case_fatality_rate     0.869               1.000      0.996
cfr_lag14              0.875               0.996      1.000
```

Figure 5. Correlation matrix showing the relationships among ICU Rate, Case Fatality Rate, new cases, and hospital admissions. The ICU Rate and CFR demonstrate a strong positive correlation (r = 0.869), indicating that periods of increased ICU occupancy were closely associated with higher fatality rates.

Mortality Metrics

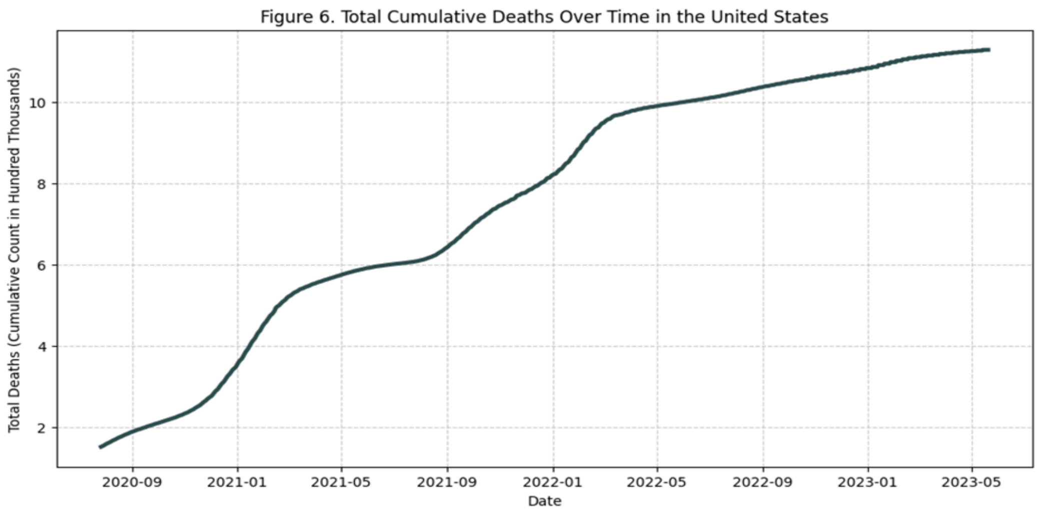
This section analyzes COVID-19 mortality trends in the United States from mid-2020 through mid-2023. Using both raw mortality indicators (new deaths, total deaths) and derived severity metrics (case fatality rate, ICU rate, hospitalization rate), we examine how mortality evolved and how different phases of the pandemic can be distinguished.

Time-Series Analysis of Mortality Trends



The plot of the 7-day average of new deaths reveals clear, distinct mortality waves throughout the pandemic:

+ **Wave 1 (Winter 2020 - 2021):** The deadliest and most prolonged mortality spike occurred between December 2020 and January 2021, peaking above 3,000 deaths per day. This aligns with the pre-vaccination environment.
+ **Wave 2 (Late 2021 - Early 2022):** A second major spike appears during the Delta variant (Sept - 2021) and Omicron variant (late 2021 - early 2022) waves. Although Omicron was less severe, its sheer transmissibility produced a significant mortality rise.
+ **Declining Trend in 2022–2023:** After early 2022, the mortality curve steadily declines. Sporadic, smaller bumps persist, but deaths remain far below earlier numbers.

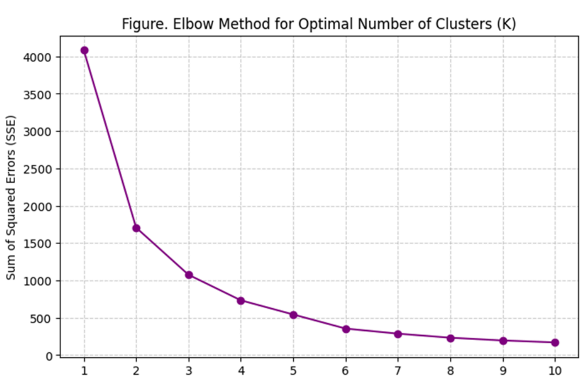


This plot shows how the total number of COVID-19 deaths in the United States **increased over time**, from mid-2020 to mid-2023. During **early 2021**, the total deaths **rapidly rose** from approx 200k to approx 500k cases, marking the deadliest period of the U.S. pandemic. Around mid-2021, the curve starts to **flatten**, indicating fewer deaths per day. This aligns with the **mass vaccination campaign**, which began in early 2021. A second sharp incline appears in **late 2021 to early 2022**, corresponding to the **Delta and Omicron variant waves**. After **early 2022**, the rate of increase becomes **much slower**. By 2023, daily death counts had **significantly decreased** and **stabilized**.

Pandemic Phase Classification Using Clustering

To better understand different mortality levels, we used K-Means clustering on four key severity and mortality indicators, which are 7-day average new deaths, case fatality rate (CFR), ICU rate and hospitalization rate. This unsupervised clustering uncovers natural groupings of pandemic days based on severity patterns rather than arbitrary calendar boundaries.

Elbow Method to Determine Optimal k



Determining the Number of Phases (k), the elbow method is used to find the optimal number of clusters (k). The plot of the Sum of Squared Errors (SSE) often shows a sharp "elbow" where adding more clusters yields diminishing returns. For this dataset, a clear elbow often appears at k=3, so I chose k=3 to define three highly interpretable phases of the pandemic.

Cluster Characteristics

After training, the model identified **three distinct mortality phases**, **the highest, mid-level, and lowest mortality rates,** each representing a unique pandemic condition:

Cluster 1 (High deaths, moderate severity): Characterized by the highest death counts but only moderate severity rates (CFR, ICU). This cluster represents the massive volume surges where the sheer number of cases drove up the deaths.

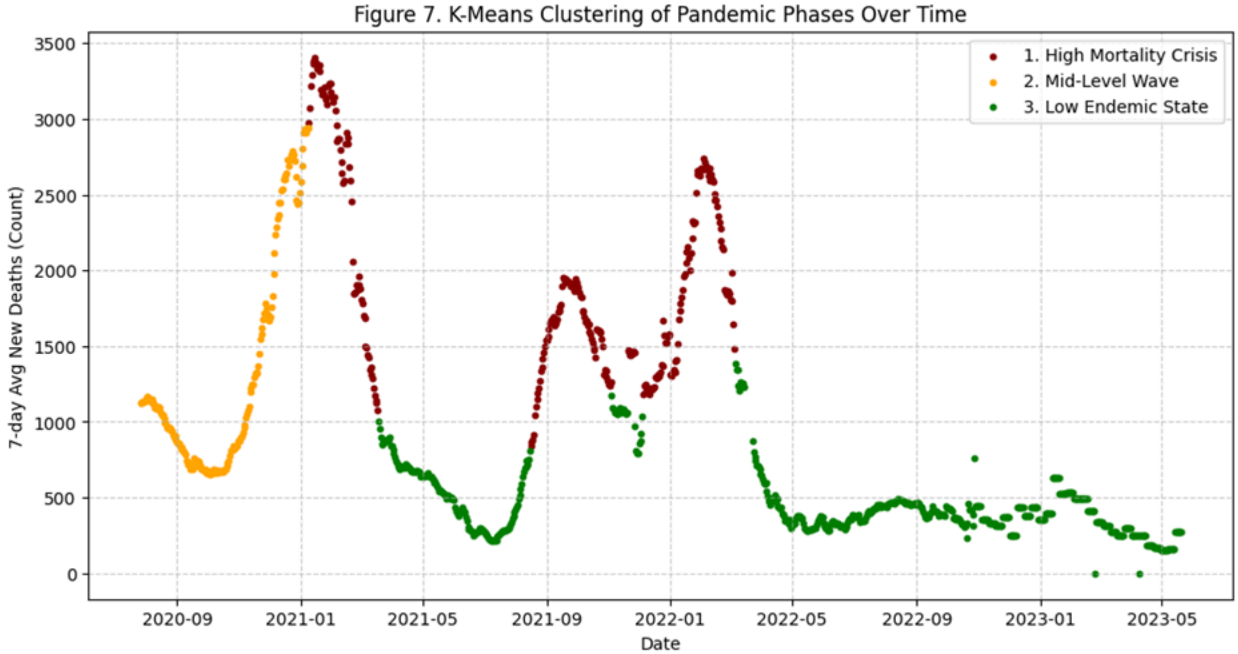
Cluster 2 (High severity rates, early pandemic): Characterized by having the highest mortality and severity rates (CFR approx 2.64%, ICU Rate approx 0.16%).

Cluster 3 (Endemic Stability State): Characterized by the lowest values across all metrics, representing the post-vaccine endemic periods.

Phase Distribution Over Time

By visualizing the clusters along the timeline (Figure 7), it shows a clear chronological segmentation:

Figure 7. K-Means Clustering of Pandemic Phases Over Time

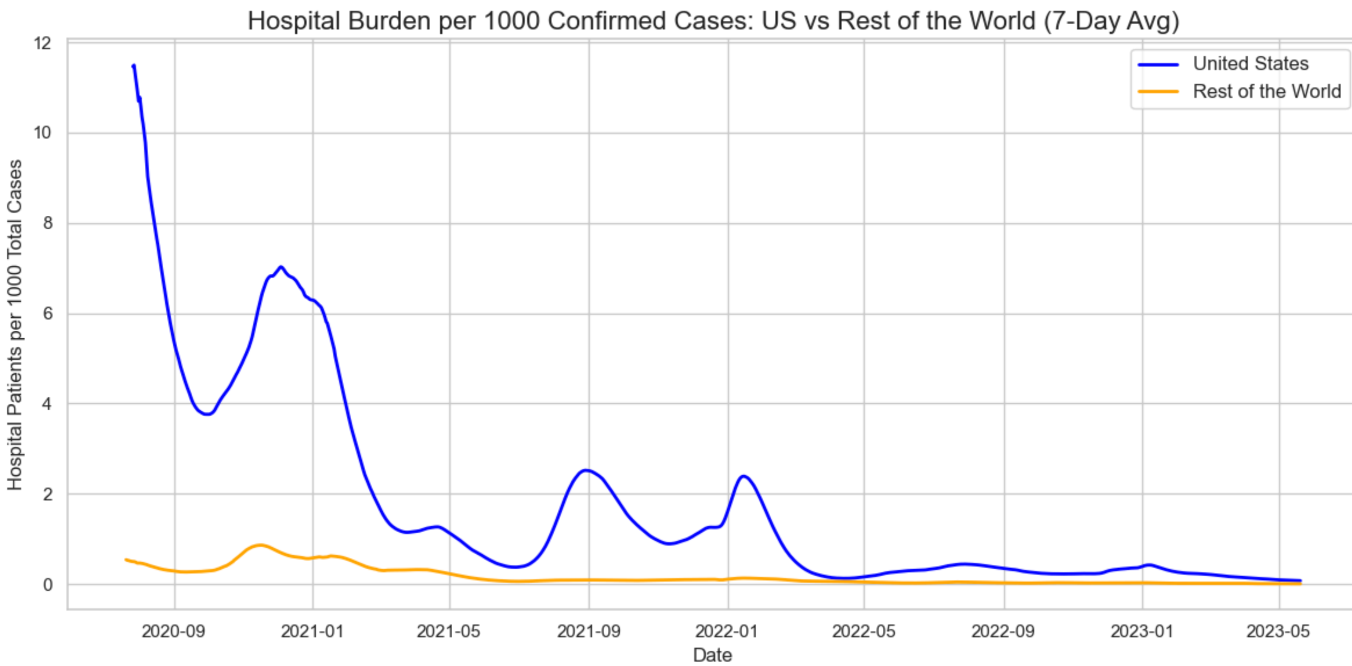


This figure shows how the three mortality phases map onto the pandemic timeline. **High-severity and high-mortality** conditions are clustered **early on**, while **milder endemic conditions** dominate the **later period**. The segmentation reflects the COVID-19 real-world epidemiological transitions driven by vaccination, immunity, and changing variants.

COVID-19 Burden Comparison

This short report summarizes the findings derived from the analysis comparing the Hospital and ICU Burden per 1000 Confirmed COVID-19 Cases in the United States (US) versus the Rest of the World (RoW). The analysis employed time-series exploration, regression, and distribution methods.

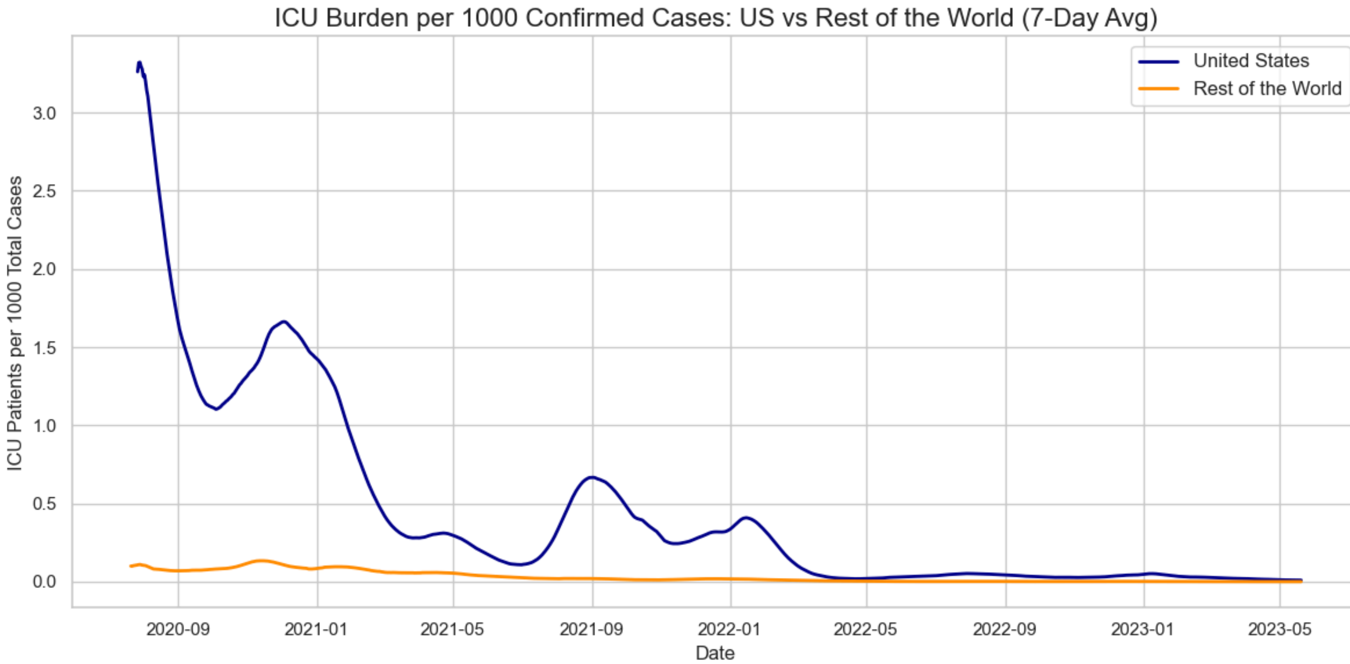
1. **Burden Rate Time Series Analysis**



The US consistently demonstrated a significantly higher Hospital Burden Rate than the RoW throughout the observed period (starting July 2020).

The US rate typically peaked around 10 to 15 hospital patients per 1000 confirmed cases, while the RoW rate remained close to 0 or below 5.

This suggests that, relative to its cumulative case count, the US health system carried a substantially heavier load of *active* hospitalized patients than the global average.



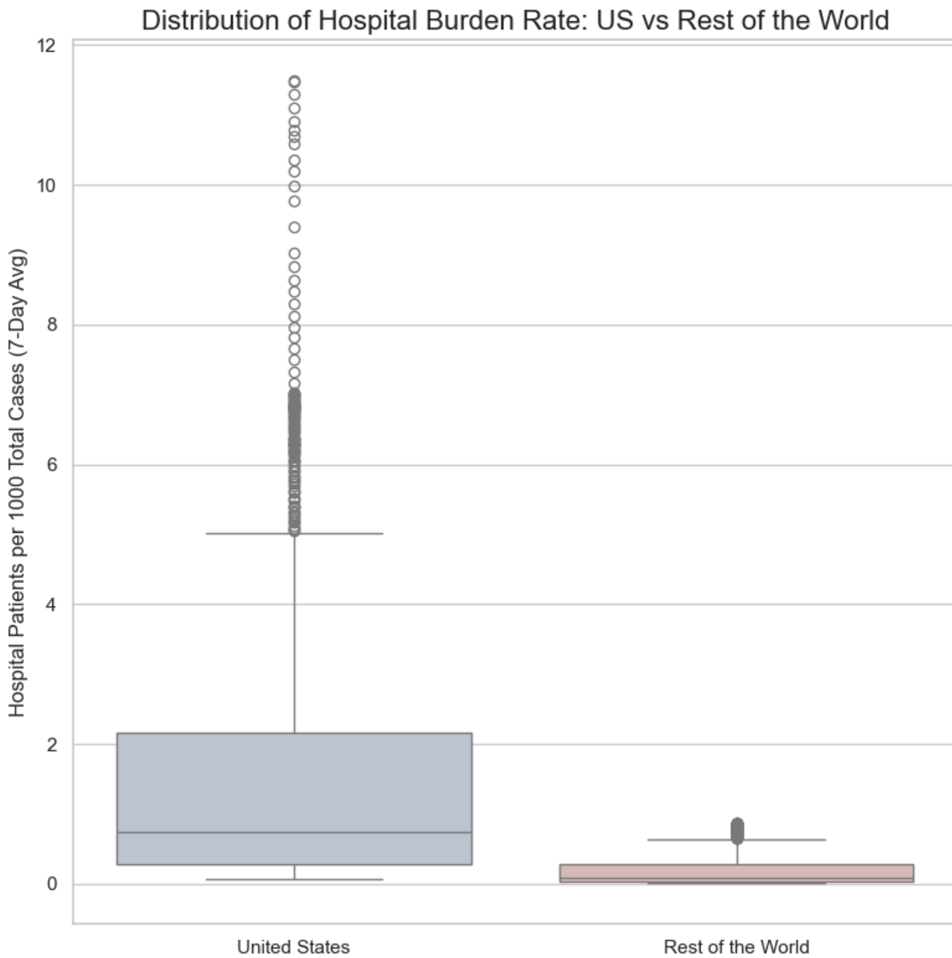
Similar to hospitalization, the US also exhibited a higher ICU Burden Rate, often peaking between 2 to 4 ICU patients per 1000 confirmed cases.

The RoW ICU burden was consistently lower, often barely registering above 0.

This confirms the initial hypothesis: the United States experienced a higher hospital and ICU burden per cumulative confirmed case than the Rest of the World.

2. **Distribution and Outlier Analysis**

The Box Plot provides a statistical overview of the 7-day average Hospital Burden Rate



Distribution Comparison:

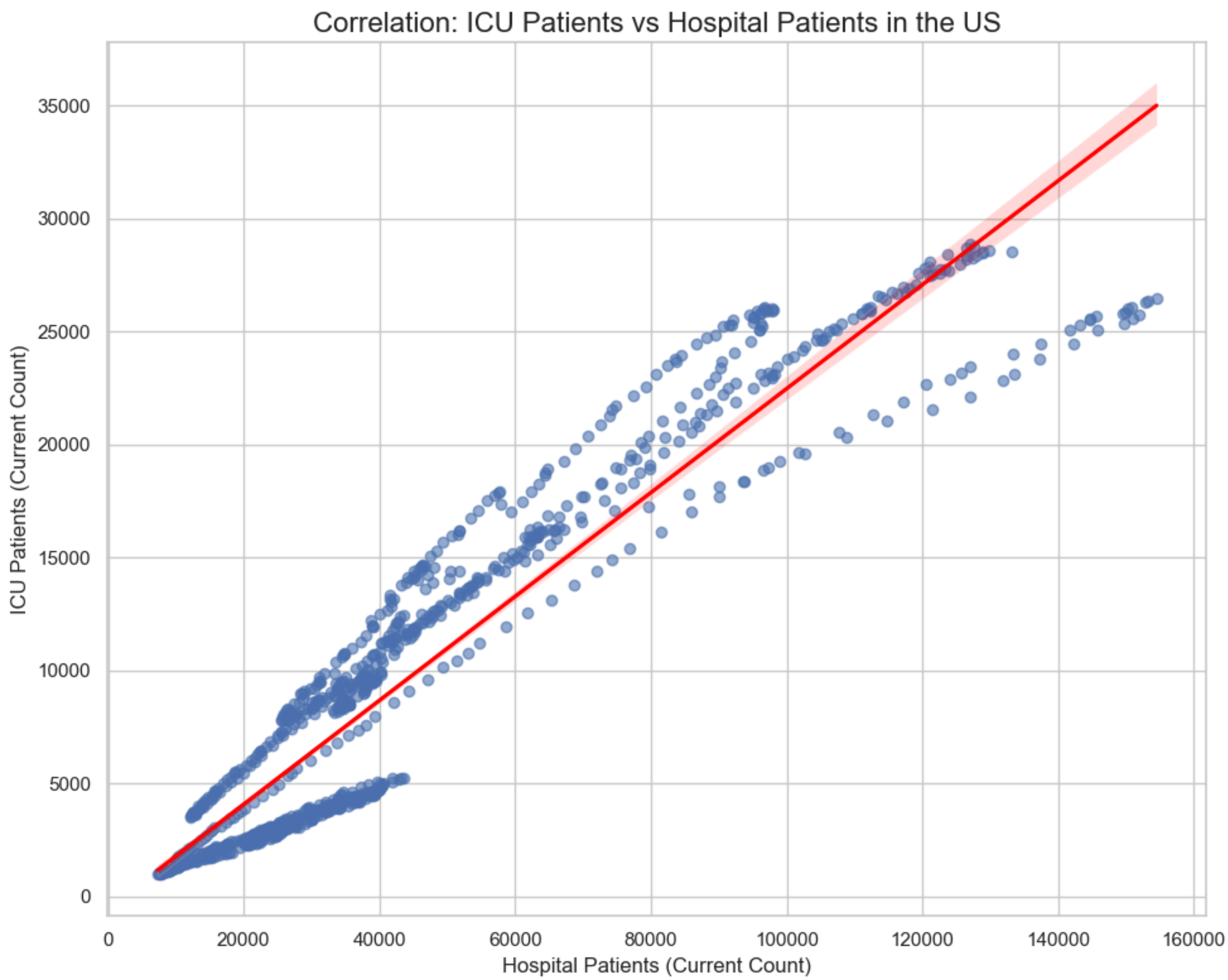
•The US distribution (box) is higher than the RoW distribution, confirming that the median and interquartile range (IQR) of the US burden rate were consistently greater.

•The median (the line inside the box) for the US is notably higher than the RoW median.

•The plot reveals outliers for the US (points above the whisker), indicating several periods where the burden rate climbed to exceptional levels, surpassing the general trend seen in the RoW. The RoW distribution is heavily skewed toward low values.

3. **Correlation Analysis (US Specific)**

The Regression Plot demonstrates the fundamental internal relationship between patient metrics in the US.



The scatter plot with a regression line shows an extremely strong, positive linear correlation between the number of current Hospital Patients and the number of current ICU Patients in the United States.

As the total number of hospital patients rises, the number of ICU patients rises proportionally. This is expected, as ICU patients are a subset of hospital patients and is a necessary validation for the quality and reliability of the US patient data.

Conclusion: Based on the analysis of the burden rate trends and distributions, the data strongly supports the conclusion: The United States experienced a substantially higher Hospital and ICU Burden per 1000 Confirmed COVID-19 Cases than the Rest of the World. This means that for every 1000 people who tested positive in the US, a larger fraction required simultaneous hospital or ICU care compared to the global average over the observed period.